

Figure 1: Overview of the AdaSolver framework. The diagram illustrates the input geometry (a car) and the resulting pressure distribution (Pa) on its surface. The color bar indicates the pressure range from -1300 to 600 Pa, with a ground truth visualization showing high pressure on the rear and low pressure on the front. The AdaSolver framework is compared against Uniform and Adaptive solvers. The Uniform solver produces an inaccurate result, the Adaptive solver produces a biased result, and the AdaSolver produces an accurate result, as indicated by the green checkmark and the label 'Accurate'.

Input points $x \in \mathbb{R}^{N \times C}$

Soft slice $W \in \mathbb{R}^{N \times M}$

Measure correction

$$c_i = \left(\frac{m_i}{\tilde{m}_i} \right)^\gamma$$

$m_i \quad \tilde{m}_i \quad c_i$

element - size

Physics tokens $T \in \mathbb{R}^{M \times C}$

Token attention

$$t_j = \frac{\sum_i c_i w_{ij} h_i}{\sum_i c_i w_{ij}}$$

Reuse slice weights

Deslice

Output fields $y \in \mathbb{R}^{N \times C}$

broadcast

The diagram illustrates the AdaSO framework for design optimization. It begins with **Geometry** and a **Physics prior Normal variation**. The geometry is converted into a point cloud S_0 (10%). This point cloud is processed by a **Lightweight head** (a neural network) to produce **Learned offline** features, specifically **Frozen features** and **Feature norm (a.u.)** (ranging from 400 to 900). The framework then branches into three refinement methods:

- Physics-Informed Density Refinement (PI)**: Uses **Normal variation** for **Adaptive Sampling** to refine **Sample points**, resulting in **PI pressure** (ranging from -1300 to 600 Pa).
- Coarse-Prediction-Guided Density Refinement (CPG)**: Uses **Field variation** for **Adaptive Sampling** to refine **Sample points**, resulting in **CPG pressure** (ranging from -1300 to 600 Pa).
- Cache-Aware Learned Refinement (CA)**: Uses a **Learned score** (ranging from 0 to 1) for **Adaptive Sampling** to refine **Sample points**, resulting in **CA pressure** (ranging from -1300 to 600 Pa). This method also tracks the **Local retained fraction** (0% to 100%).

The refined pressure fields are used to inform the **Design** process, which involves parameters $C_d, C_l, \dots$. The design is evaluated against the **Design objective** (Optimization time) and **Prediction** (Step). The design objective plot shows that AdaSO (green solid line) is faster and better than uniform sampling (black dashed line) and GT (grey dotted line). The prediction plot shows that AdaSO (green solid line) has a smaller error than uniform sampling (black dashed line) and GT (grey dotted line).